GITAA Pvt.Ltd.
Transforming careers

Course Outline

Week 1 — Introduction to AI and Data Science

Modules

- Overview of Artificial Intelligence (AI) and Data Science
- Introduction to AI, Machine Learning (ML), and Deep Learning (DL)
- Applications of AI and Data Science across various industries

Week 2 — Python Basics and Sequence Data Types

Modules: Python Fundamentals

- Basic Python Syntax
- Variables and Data Types
- Type Casting
- Python Operators and Order of Operations
- Control Structures

Modules: Sequence Data Types

- Introduction to Lists, Tuples, Dictionaries, Strings, Ranges, and Sets
- Indexing and Slicing
- Common Methods & Functions
- Specific Operations on each Sequence Type
- Introduction to NumPy

Week 3 — Descriptive Statistics

Modules

- Building Blocks of Data Science
- Introduction to Descriptive Statistics
- Measures of Central Tendency (Mean, Median, Mode)
- Measures of Dispersion (Range, Variance, Standard Deviation)
- Measures of Association Between Variables (Correlation, Covariance)

Week 4 — Data Handling with Pandas

Modules

- Introduction to Pandas DataFrames
- Reading and Writing Files (CSV, Tab-delimited, Excel)
- Exploratory Data Analysis (EDA)
- Data Preparation and Preprocessing

Hands-on Case Study: Data Cleaning & Preprocessing

- Understanding Data Sources and Problem Statement
- Handling Missing Data & Outliers
- Removing Duplicates and Inconsistencies

Week 5 — Data Visualization with Python

Libraries Used

- Matplotlib
- Seaborn

Modules

- Scatter Plots
- Bar Charts
- Histograms
- Box Plots
- Pair Plots

Week 6 — Introduction to Tableau

Modules

- Overview of Tableau Interface
- Creating and Formatting Charts
- Designing Dashboards and Layouts

Week 7 — Inferential Statistics

Modules

- Populations, Samples, and Sampling Distributions
- Confidence Intervals and Hypothesis Testing

- Practical Exercises for Statistical Inference

Week 8 — Machine Learning Fundamentals: Linear Regression

Modules

- What is Machine Learning?
- Introduction to Linear Regression
- Model Building and Evaluation Metrics

Hands-on Session

- Implementing Linear Regression in Python

Week 9 — Decision Trees & Random Forests

Modules

- Introduction to Decision Trees
- Introduction to Random Forest Algorithms

Hands-on Session

- Building and Evaluating Decision Tree & Random Forest Models

Week 10 — Logistic Regression, KNN, and Clustering

Modules: Supervised Learning

- Introduction to Logistic Regression
- Introduction to K-Nearest Neighbors (KNN)

Modules: Unsupervised Learning

- K-Means Clustering
- Hierarchical Clustering

Hands-on Session

- Implementing Logistic Regression, KNN, and Clustering Models

Week 11 — Dimensionality Reduction & PCA

Modules

- Introduction to Dimensionality Reduction Methods

- Principal Component Analysis (PCA) and Its Variants

Week 12 — Advanced ML & Deep Learning

Modules

- Support Vector Machines (SVM)
- Introduction to Deep Learning
- Basics of Neural Networks

Week 13 — Generative AI & Prompt Engineering

Modules: Introduction to Generative AI Tools

- Overview of ChatGPT, Gemini, DeepSeek, and others
- Capabilities and Limitations of Generative AI Tools
- Ethical and Responsible Usage

Modules: Effective Prompt Engineering

- Writing Clear, Specific Prompts for Programming Help
- Prompts for Explaining Python Errors and Debugging Suggestions
- Generating Code Snippets (File Operations, Loops, Conditionals, etc.)
- Dos and Don’ts in Prompt Formulation

Week 14 — Automation with AI

Modules: AI-Assisted Data Cleaning

- Handling Missing Values
- Removing Duplicates
- Data Type Conversions